higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANICAL DRAWING AND DESIGN N5

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This marking guideline consists of 9 pages.

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QUESTION 1

1.1 $Effective\ length\ of\ weld = Overall\ length\ of\ weld - 2 \times leg$ ¿¿

$l = 150\ mm - 2 \times 10\ mm$

$l = 150\ mm - 20\ mm$

$l = 130\ mm$✓

$Throat$ ¿ $0{,}707 \times w$

$= 0{,}707 \times 10\ mm$

$= 7{,}07\ mm$✓

$Throat\ area = Throat$ ¿ $\times\ effective\ length$

$= 7{,}07\ mm \times 130\ mm$

$= 919{,}1\ mm^2$✓

$Effective\ length\ of\ weld = The\ full\ 150\ mm\ Length\ is\ taken$

$Throat\ area = Throat$ ¿ $\times\ effective\ length$

$= 7{,}07\ mm \times 150\ mm$

$= 1060{,}5\ mm^2$✓✓ (5)

1.2 $$Tensile\ stress = \frac{Tensile\ load}{Cross\text{-}sectional\ area}$$

$$\sigma_t = \frac{P}{t_{throat} \times l}$$

$$l = \frac{P}{t_{throat} \times \sigma_t}$$✓

¿ $$\frac{44{,}5 \times 10^3\ N}{0{,}006\ m \times 138 \times 10^6\ N\,m^2}$$✓

$= 0{,}0537\ m$✓

$say\ l = 54\ mm$✓✓ (5)

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QUESTION 2

2.1 $$Tensile\ \ stress = \frac{Tensile\ \ load}{Cross\text{-}sectional\ \ area\ \ of\ \ plate}$$

¿ $$\frac{P}{(p-d)\times t}$$

¿ $$\frac{30\times10^{3}\,N}{(0{,}05\ \ m-0{,}02\ \ m)\times0{,}012\,m}$$

$$=83{,}3\times10^{6}\,N/m^{2}$$

$$=83{,}3\ \ MPa$$✓✓ (3)

2.2 $$Shear\ \ stress = \frac{Shear\ \ load}{Cross\text{-}sectional\ \ area\ \ of\ \ rivet}$$

¿ $$\frac{\frac{P}{\pi d^{2}}}{4}$$

¿ $$\frac{\frac{30\times10^{3}\,N}{\pi\times(0{,}02\ \ m)^{2}}}{4}$$

$$=95{,}5\times10^{6}\,N/m^{2}$$✓
$$=99{,}5\ \ MPa$$✓✓ (3)

2.3 $$Crushing\ \ stress = \frac{Crushing\ \ load}{Projected\ \ area\ \ of\ \ contact}$$

¿ $$\frac{P}{d\times t}$$

¿ $$\frac{30\times10^{3}\,N}{0{,}02\ \ m\times0{,}012\ \ m}$$

$$=125\times10^{6}\,N/m^{2}$$✓

$$=125\,MPa$$✓✓ (3)

2.4 $$Ultimate\ \ shear\ \ load = Ultimate\ \ shear\ \ stress\ \times cross\text{-}sectional\ \ area\ \ of\ \ rivet$$

$$=\tau\times\frac{\pi d^{2}}{4}$$

$$=384\times10^{6}\,N/m^{2}\times\frac{\pi\times(0{,}02)^{2}}{4}$$

$$=120{,}64\times10^{3}\,N$$✓ (3)

$= 120{,}64 \ \ kN$ ✓✓

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QUESTION 3

Data:

Driving pulley: $D_1 = 320\ \ mm;\ \ N_1 = 680\ \ r/min$
Driven pulley: $D_2 = 600\ \ mm$
$\mu = 0{,}4;\ C = 900\ \ mm;\ \ T_1 = 7\ kN;\ \ t = 10\ \ mm;\ \ \sigma_t = 4{,}5\ \ MPa$

3.1 Contact angle

$$\sin\beta = \frac{D_1 + D_2}{2_c}$$

$$\therefore \beta = \sin^{-1}\frac{0{,}32 + 0{,}6}{2(0{,}9)} = 30{,}738\,° \checkmark$$

$$\therefore \theta = 180\,° + 2\,\beta = 10\,° + 2(30{,}738\,°) = 241{,}476\,° \checkmark\checkmark \quad (3)$$

3.2 Width of belt

$$T_1 = wt\sigma_t$$

$$\therefore w = \frac{T_1}{t\sigma_t} = \frac{7 \times 10^2}{(0{,}01)(4{,}5 \times 10^6)} = 0{,}155\ \ 555\ \ m$$

Say w = 160 mm✓✓ (2)

3.3 Total length of belt

$$L = \frac{1}{2}\pi(D_1 + D_2 + 2_t) + \frac{(D_1 + D_2 + 2_t)^2}{4C} + 2C$$

$$L = \frac{1}{2}\pi[0{,}32 + 0{,}6 + 2(0{,}01)] + \frac{[0{,}32 + 0{,}6 + 2(0{,}01)]^2}{4(0{,}9)} + 2(0{,}9)\ \checkmark$$

$$L = 1{,}476\ \ 6 + 0{,}245\ \ 44 + 1{,}8$$
$$L = 3{,}522\ \ m \checkmark\checkmark \quad (3)$$

3.4 Power transmitted by belt

$$tension\ \ ratio: \theta = \frac{241{,}476\,°}{57{,}3} = 4{,}214\ \ radians\ \checkmark$$

$$\frac{T_1}{T_2} = e^{\mu\theta} = e^{(0{,}4)(4{,}214)} \checkmark$$

$$\therefore T_1 = 5{,}396\,T_2$$

$$For\ \ T_2: \quad T_1 = 5{,}3962\,T_2$$

$$\therefore T_2 = \frac{T_1}{5{,}396} = \frac{7\ \ 000}{5{,}396} = 1297{,}257\ \ N \checkmark\checkmark \quad (7)$$

$for\ \ power:$
$P=(T_1-T_2)v=(7\ \ 000-1\ \ 297{,}257)(11{,}75)=67\ \ 007{,}23\ \ W$ ✓
$\therefore P=67\ \ kW$✓✓

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QUESTION 4

4.1 Allowable stresses

$$Shear\ \ stress=\frac{417\times10^6\,N/m^2}{5}=83{,}4\ \ MPa$$✓

$$Tensile\ \ stress=\frac{460\times10^6\,N/m^2}{5}=92\ \ MPa$$✓

$$Crushing\ \ stress=\frac{834\times10^6\,N/m^2}{5}=166{,}8\ \ MPa$$✓

(3)

4.2 Diameter of pierced rod d_1

$$d_1=\sqrt{\frac{4\times(\sigma_c\times\sigma_t)}{\pi(\sigma_c\times\sigma_t)}}$$

$$¿\sqrt{\frac{4\times60\times10^3(166{,}8\times10^6+92\times10^6)}{\pi(166{,}8\times10^6+92\times10^6)}}$$

$$¿\sqrt{1{,}2884\times10^{-3}}$$
$d_1=0{,}0359\ \ m$
Say 40 mm✓✓

(2)

4.3 Breadth of cotter
$Shear\ \ load\in cotter=crushing\ \ load\in cotter\ \ against\ \ rod$
$\tau\times2\times t\times b=\sigma_c\times d_1\times t$

$$b=\frac{\sigma_c\times d_1}{\tau\times2}$$

$$¿\frac{166{,}8\ \times\ 10^6\,N/m^2\ \times\ 0{,}04\ \ m}{83{,}4\ \times\ 10^6\,N/m^2\ \times\ 2}$$

$=0{,}04\ \ m$
$b=40\ \ mm$✓✓

(2)

4.4 Cotter thickness
$Tensile\ \ load\in pierced\ \ rod=Crushing\ \ load\in cotter\ \ against\ \ rod$

$$\left(\frac{\pi d_1^2}{4}-d_1\times t\right)=\sigma_t\times d_1\times t$$

(5)

$$t = \frac{\sigma_t \times \pi \times d_1}{4\left(\sigma_c \times \sigma_t\right)} \checkmark$$

¿ $\frac{92 \times 10^6 N/m^2 \times \pi \times 0{,}04\ m}{4\left(166{,}8 \times 10^6 N/m^2 + 92 \times 10^6 N/m^2\right)}$ ✓

$t = 0{,}0111$ ✓

Say 11 mm ✓✓

4.5 Diameter of pierced rod d_1

$$d_1 = \sqrt{\frac{4 \times (\sigma_c \times \sigma_t)}{\pi (\sigma_c \times \sigma_t)}}$$

$$\therefore \sqrt{\frac{4 \times 60 \times 10^3 (166{,}8 \times 10^6 + 92 \times 10^6)}{\pi (166{,}8 \times 10^6 + 92 \times 10^6)}}$$

$$\therefore \sqrt{1{,}2884 \times 10^{-3}}$$

$$d_1 = 0{,}0359 \ \ m \checkmark$$

Say 40 mm✓✓ (3)
[15]

QUESTION 5

5.1 Torque transmitted

$$Power = \frac{2 \pi T_{mean} N}{60}$$

$$T_{mean} = \frac{Power \times 60}{2 \times \pi \times N} \ \checkmark$$

$$\therefore \frac{125 \times 10^3 \times 60}{2 \times \pi \times 440}$$

$$= 2713 \ \ Nm \checkmark\checkmark$$ (3)

5.2 Shear stress in the shaft
Assume
$T_{mean} = T_{max}$

$$T = \frac{\pi d_s^3}{16} \ \tau$$

$$\tau = \frac{T \times 16}{\pi d_s^3} \checkmark$$

$$\therefore \frac{2713 \ \ Nm \times 16}{\pi \times (0{,}07 \ \ m)^2}$$

$$= 40{,}28 \ \ MPa \checkmark\checkmark$$ (3)
[6]

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QUESTION 6

6.1 $$T = \frac{Power \times 6}{2\pi N}$$

$$¿ \frac{18 \times 10^3 \times 60 \times 6}{2 \times \pi \times 1000 \ \ r/min}$$

$$T = 1031,13 \quad Nm ✓✓$$ (2)

6.2 $$Torque = Force \times Perpendicular \ \ distance$$

$$T = P \times R$$

$$1031,13 \ = P \times \frac{0,152 \ \ m}{2}$$

$$P = \frac{1031,13 \times 2}{0,152}$$

$$P = 13478,82 \ \ N ✓✓$$ (2)

6.3 $$Force \ \ on \ \ each \ \ pin = \frac{Total \ \ force}{Number \ \ of \ \ pins}$$

$$¿ \frac{13478.82}{6}$$

$$= 2246 \ \ N ✓✓$$ (2)

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QUESTION 7

Data:

$$X = 350 \ \ mm ; Y = 310 \ \ mm ; \ \varphi = 1,5 \ MN/m^2 ; \ \sigma_t = 40 \ \ MPa ; M \ 22$$

7.1 For effective area of jointing material

$$a = X + 2t + 2d + 2 \ clearance = 350 + 2(23) + 2(22) + 2(4) = 448 \ \ mm ✓$$
$$b = Z + 2t + 2d + 2 \ clearance = 301 + 2(23) + 2(22) + 2(4) = 408 \ \ mm ✓$$
$$A = a \times b = (0,448)(0,408) = 0,182 \ \ 784 \ \ m^2 ✓✓$$ (5)

7.2 For force extended on cover

$$F = \varphi \ \ A = (1,5 \times 10^6)(0,182 \ \ 784) = 274,176 \ \ kN ✓✓$$ (2)

7.3 For number of bolts

$$F = F_? = 0,8 \, n \left(\frac{1}{4}\pi d^2\right) \sigma_t$$ (3)

$$\therefore n = \frac{4F}{0{,}8\pi d^2 \sigma_t} = \frac{4\left(274{,}176 \times 10^3\right)}{0{,}8\pi\left(0{,}022\right)^2\left(40 \times 10^6\right)} = 22{,}539 \;\; 5 = 24 \;\; bolts \checkmark\checkmark$$

7.4 For pitch of bolts

$$L_x = \frac{X + 2t + 3d}{5} = \frac{350 + 2(23) + 3(22)}{5} = 92{,}4 \ mm ✓✓$$

$$L_y = \frac{Y + 2t + 3d}{4} = \frac{310 + 2(23) + 3(22)}{4} = 105{,}5 \ mm ✓✓$$ (4)

7.5 For steam tightness

$1{,}5 \ MN/m^2 \ is \ above \ 1{,}125 \ MPa \ \therefore \ pitch \ must \ fall \ between \ 3d \ \wedge \ 5d$

$3d = 3(22) = 66 \ mm \quad \wedge \quad 5d = 5(22) = 110 \ mm ✓$

Yes, steam tight✓ (2)

7.6 For total length and breadth of cover

$Length = X + 2t + 6d = 350 + 2(23) + 6(22) = 528 \ mm ✓✓$

$Breadth = Y + 2t + 6d = 310 + 2(23) + 6(22) = 488 \ mm ✓✓$ (4)

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QUESTION 8

Length required to resist shear

$Torque \ on \ shaft = Torque \ on \ key$

$$\tau_{shaft} = T_{key} \times W \times l \times \frac{d_s}{2}$$

$$680 \ Nm = 55 \times 10^6 \ N/m^2 \times 0{,}013 \times l \times \frac{0{,}038 \ m}{2}$$

$$l = \frac{680 \ Nm \times 2}{55 \times 10^6 \ N/m^2 \times 0{,}013 \times 0{,}038 \ m}$$

$= 0{,}05 \ m ✓$

$l = 50 \ mm ✓✓$

Length required to resist compression

$Torque \ on \ shaft = Torque \ on \ key$

$$\tau_{shaft} = Q_{ckey} \times \frac{1}{2} \times l \times \frac{d_s}{2}$$

$$680 \ Nm = 110 \times 10^6 \ N/m^2 \times \frac{0{,}01 \ m}{2} \times l \times \frac{0{,}038 \ m}{2}$$

$$l = \frac{680 \ Nm \times 2 \times 2}{110 \times 10^6 \ N/m^2 \times 0{,}01 \times 0{,}083 \ m}$$

[6]

$= 0{,}065 \ \ m$ ✓
$l = 65 \ \ mm$ ✓✓

QUESTION 9

9.1 $Total\ load\ on\ bearings\ (Q) = 10\ 160\ kg \times 9{,}61\ m/s^2$
$= 99669{,}6\ newtons$ ✓✓ (2)

9.2 $Total\ friction\ force\ (F) = \mu \times Q$
$= 0{,}075 \times 99669{,}6$
$= 7475{,}22\ newtons$ ✓✓ (2)

9.3 $Distance\ moved \in metres\ per\ minute = \pi \times d \times N$
$= \pi \times 0{,}3 \times 70$
$= 65{,}97\ metres\ per\ minute$ ✓✓ (2)

9.4
$Work\ done\ per\ minute = Friction\ force \times distance\ moved\ per\ minute = 7475{,}22\ N$
$= 493\ 140{,}26\ Nm\ per\ minute$ ✓✓ (2)

9.5 $Power\ absorbed\ by\ friction = frictionforce \times V$
$= \mu \times Q \times V$

$= 7\ 475{,}22\ N \times \frac{65{,}97}{60}\ m/s$

$= 7\ 097{,}72\ watts$
$= 8\ 219\ kW$ ✓✓ (2)
[10]

TOTAL: 100

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